SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem *(2)*	Application of Engineering Concepts *(2.5)*	Solution Design *(2.5)*	Use of Tools *(2)*	Analysis *(1)*	*Total(10)*
Weightage	20%	25%	25%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I G Madhulatha(192311295)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
G Madhulatha

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:
Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Q2. Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:
Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:
Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Q4 . Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:
Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:
Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

INDUSTRY PROBLEM-SOLVING TASK

ANSWERS

Q1. Dataset: Supermarket Purchases

Question
Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Answer
1. Problem Identification

A supermarket has customers with different purchasing behaviors. Some customers visit the store frequently but spend less per visit, while others visit less frequently but spend more.
The given features are:

- **Visits/Month**
- **Average Spend (₹)**
- **Items Bought**

The objective is to divide customers into groups having similar purchasing behavior.

2. Suitable Clustering Technique

K-Means clustering is suitable for this problem because:

- The dataset contains numerical features.
- Customers can be divided into a predefined number of groups.
- It is simple and efficient.
- Cluster centers can be easily interpreted for marketing decisions.

Before applying K-Means, the features should preferably be **normalized/standardized**, especially because average spending has values in thousands while visits and items are much smaller.

3. Dataset Analysis

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2,000	15
C2	4	5,000	8
C3	12	1,800	18
C4	3	6,000	6
C5	11	2,100	16

There are two obvious behavioral patterns.

4. Possible Cluster Formation

Cluster 1 – Frequent, Low-Spending Customers

Customers:
C1, C3, C5
Characteristics:

- Visits are high: 10–12 visits/month.
- Average spending is relatively low: ₹1,800–₹2,100.
- Items bought are high: 15–18.

These customers visit frequently and buy many items but have relatively low spending per visit.

Marketing strategy:

- Give loyalty rewards.
- Offer premium product recommendations.
- Provide "Buy More, Save More" offers.
- Give coupons for higher-value products.
- Encourage customers to increase average basket value.

Cluster 2 – Infrequent, High-Spending Customers

Customers:
C2, C4
Characteristics:

- Visits are low: 3–4 visits/month.
- Average spending is high: ₹5,000–₹6,000.
- Items bought are lower: 6–8.

These customers make fewer visits but spend significantly more when they purchase.

Marketing strategy:

- Provide personalized premium offers.
- Send reminders to encourage repeat visits.
- Offer loyalty memberships.
- Recommend complementary products.
- Provide special discounts for frequent visits.

5. Clustering Process
The K-Means process can be:
Step 1: Select features:
Visits/Month + Average Spend + Items Bought
Step 2: Normalize the features.
Step 3: Select K, for example, **K = 2**.
Step 4: Initialize two centroids.
Step 5: Calculate the distance of each customer from the centroids.
Step 6: Assign customers to the nearest centroid.
Step 7: Recalculate the centroids.
Step 8: Repeat until the clusters become stable.
6. How Clustering Supports Targeted Marketing
Clustering allows the supermarket to avoid using the same marketing strategy for every customer.

Customer Segment	Behavior	Marketing Strategy
Frequent, low-spending	Many visits, lower spending	Basket-size offers, loyalty rewards
Infrequent, high-spending	Few visits, high spending	Repeat-visit offers, premium promotions

7. Business Benefits
Clustering helps the supermarket:

- Increase customer engagement.
- Improve customer loyalty.
- Increase average order value.
- Provide personalized offers.
- Reduce unnecessary advertising.
- Identify valuable customer segments.
- Improve overall sales.

Conclusion
K-Means clustering with normalized purchasing features can effectively segment supermarket customers into behavior-based groups. In this dataset, **C1, C3, and C5** form a frequent-shopping group, while **C2 and C4** form an infrequent but high-spending group. These segments allow the supermarket to design **targeted promotions, loyalty programs, and personalized recommendations**, ultimately improving sales and customer retention.

Q2. Dataset: Credit Card Transactions

Question
Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Answer
1. Problem Identification
Credit card fraud may occur when a transaction is significantly different from normal transaction behavior.
The dataset contains:

- Transaction amount
- Location
- Transaction time

The objective is to identify transactions that behave differently from the normal transaction pattern.
2. Suitable Clustering Approach
A suitable solution is **K-Means clustering combined with anomaly detection**.
Alternatively, hierarchical clustering or DBSCAN can be used. For this small dataset, K-Means provides a simple demonstration.
3. Dataset

Transaction	Amount (₹)	Location	Time
T1	200	Chennai	10
T2	5,000	Delhi	2
T3	150	Chennai	11
T4	7,000	Mumbai	1
T5	180	Chennai	12

4. Convert Categorical Data
Location is a categorical variable.
It can be converted using **one-hot encoding**.
For example:

Transaction	Amount	Time	Chennai	Delhi	Mumbai
T1	200	10	1	0	0
T2	5000	2	0	1	0

T3	150	11	1	0	0
T4	7000	1	0	0	1
T5	180	12	1	0	0

The numerical variables should also be standardized.

5. Identify Normal Behavior

Transactions:

- T1 → ₹200, Chennai, 10 hrs
- T3 → ₹150, Chennai, 11 hrs
- T5 → ₹180, Chennai, 12 hrs

These transactions have similar:

- Low amounts
- Chennai location
- Daytime transaction times

Therefore:

T1, T3, T5 → Normal transaction cluster

6. Identify Unusual Transactions

T2

- Amount = ₹5,000
- Location = Delhi
- Time = 2 hrs

This is significantly different from the normal cluster.

T4

- Amount = ₹7,000
- Location = Mumbai
- Time = 1 hr

This is also significantly different.

Therefore:

T2 and T4 are potential anomalies.

However, it is important to note that clustering identifies **unusual behavior, not confirmed fraud**. A real fraud system would require additional information such as the customer's normal location, transaction history, device information, merchant details, and authentication results.

7. Clustering-Based Fraud Detection Process

Transaction Data

↓

Data Preprocessing

↓

Encode Location

↓

Normalize Amount and Time

↓

Apply Clustering

↓

Calculate Distance from Cluster Centroid

↓

Identify Distant Transactions

↓

Flag as Potential Anomalies

↓

Further Fraud Verification

8. Fraud Detection Logic

If a transaction is:

- Very far from its assigned cluster center, or
- Belongs to a very small/isolated cluster,

then it can be flagged for investigation.

For example:

Transaction	Possible Interpretation
T1	Normal

T3	Normal
T5	Normal
T2	Potential anomaly
T4	Potential anomaly

9. Benefits
This solution can help banks:

- Detect unusual transactions.
- Identify high-value abnormal purchases.
- Detect transactions at unusual times.
- Identify unusual geographic behavior.
- Reduce manual transaction monitoring.
- Trigger additional verification.
- Improve fraud prevention.

Conclusion
A **clustering + anomaly detection approach** can identify transactions that significantly differ from normal transaction behavior. In the given dataset, **T1, T3, and T5** represent a common low-value Chennai daytime pattern, while **T2 and T4** are potential anomalies due to their high amounts, different locations, and unusual transaction times. These transactions should be **flagged for further verification rather than automatically declared fraudulent**.

Q3. Dataset: Telecom Usage

Question
Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Answer
1. Problem Identification
Customer churn occurs when telecom users stop using a service or switch to another provider.
A telecom company can use clustering to identify groups of users with similar usage patterns and determine which groups may be at greater risk of leaving.
2. Suitable Clustering Technique
K-Means clustering is suitable because the dataset contains numerical behavioral features:

- Call duration
- Data usage
- Recharge amount

The data should be normalized before clustering.
3. Dataset

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

4. Identify User Patterns
High-Usage Group
Users:
U1, U3, U5
They have:

- High call duration.
- High data usage.
- High recharge amounts.

These users are valuable customers because they use telecom services extensively.
Possible strategy:

- Premium plans
- Additional data benefits
- Loyalty rewards
- Personalized offers

Low-Usage Group
Users:
U2, U4
They have:

- Low call duration.
- Low data usage.
- Low recharge values.

These users may have a higher risk of becoming inactive, although **low usage alone does not prove churn**.
Possible strategy:

- Low-cost plans
- Customized recharge offers
- Extra data incentives
- Engagement campaigns

5. Clustering Process
The process is:
Collect telecom usage data
↓
Normalize features
↓
Apply K-Means
↓
Create customer clusters
↓
Analyze usage characteristics of each cluster
↓
Identify potentially at-risk groups
↓
Apply retention strategies
6. Possible Clusters

Cluster	Users	Behavior	Possible Action
High usage	U1, U3, U5	High calls, data and recharge	Premium loyalty programs
Low usage	U2, U4	Low calls, data and recharge	Re-engagement offers

If historical churn information is available, the company can calculate the **churn rate of each cluster**. A cluster with a high historical churn rate can be classified as a churn-prone segment.
7. How Clustering Helps Churn Detection
Clustering itself does not directly predict whether a user will leave.
Instead, it creates meaningful customer segments.
For example:

- Cluster A → High usage + high loyalty
- Cluster B → Moderate usage + declining activity
- Cluster C → Low usage + low recharge

If historical data shows that Cluster C has a high percentage of customers who previously left, then Cluster C can be considered **churn-prone**.
8. Retention Strategies
For high-value users

- Premium customer support
- Loyalty rewards
- Personalized plans
- Extra data benefits

For low-usage users

- Affordable recharge packages
- Free data trials
- Personalized discounts
- Re-engagement messages

9. Business Benefits
Clustering helps the telecom company:

- Identify customer segments.

- Understand usage behavior.
- Identify potentially risky groups.
- Personalize offers.
- Reduce customer churn.
- Improve customer satisfaction.
- Increase customer lifetime value.

Conclusion
K-Means clustering can divide telecom users according to their call duration, data usage, and recharge behavior. In the given dataset, **U1, U3, and U5** form a high-usage group, while **U2 and U4** form a low-usage group. By combining these clusters with historical churn information, the telecom company can identify **churn-prone segments and apply targeted retention strategies**.

Q4. Dataset: E-Commerce Behavior

Question
Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Answer
1. Problem Identification
An e-commerce platform has users with different browsing and purchasing behaviors.
The available features are:

- Product Views
- Clicks
- Purchases

The objective is to group users with similar behavior and use those groups to provide personalized recommendations.
2. Suitable Clustering Technique
K-Means clustering is a suitable technique.
It is appropriate because:

- The features are numerical.
- The dataset is small and structured.
- Similar users can be grouped using behavioral characteristics.
- Cluster information can be used to personalize recommendations.

The features should be normalized before applying K-Means.
3. Dataset

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

4. Analyze User Similarity
Users **U1, U3 and U5** have:

- High product views.
- High clicks.
- High purchases.

Therefore, they can form a **high-engagement/high-purchase cluster**.
Users **U2 and U4** have:

- Lower product views.
- Lower clicks.
- Lower purchases.

Therefore, they can form a **low-engagement/low-purchase cluster**.
5. Possible Clusters

Cluster	Users	Characteristics
Cluster 1	U1, U3, U5	High views, clicks and purchases
Cluster 2	U2, U4	Low views, clicks and purchases

6. Clustering Process
Collect user behavior
↓

Select views, clicks and purchases
↓
Normalize data
↓
Select K
↓
Apply K-Means
↓
Assign users to clusters
↓
Analyze product preferences within clusters
↓
Generate personalized recommendations

7. How Clustering Improves Recommendation Systems

Suppose users in one cluster frequently purchase electronics.
The system can recommend:

- Mobile phones
- Laptops
- Headphones
- Smart devices

Similarly, if another cluster frequently interacts with clothing products, the system can recommend:

- Shirts
- Shoes
- Accessories
- Fashion products

Thus, the recommendation system does not treat every user equally.

8. Example

Suppose U1, U3 and U5 belong to a high-engagement cluster.
The recommendation system can:

- Recommend similar products.
- Display premium products.
- Provide personalized discounts.
- Recommend products frequently purchased by similar users.

For U2 and U4:

- Show popular products.
- Provide introductory discounts.
- Recommend products based on their limited interactions.
- Encourage further browsing.

9. Benefits

Clustering can improve:

- **Personalization**
  Recommendations are based on similar user behavior.
- **Customer engagement**
  Relevant products encourage users to interact more.
- **Conversion rate**
  Better recommendations can increase purchases.
- **Customer satisfaction**
  Users receive more relevant products.
- **Marketing efficiency**
  Promotional campaigns can target specific customer segments.
- **Cross-selling**
  Related products can be recommended.

Conclusion

K-Means clustering can group e-commerce users according to product views, clicks, and purchases. In this dataset, **U1, U3, and U5** represent highly engaged users, while **U2 and U4** represent lower-engagement users. These clusters can be integrated into a recommendation system to provide **personalized products, offers, and advertisements**, improving customer engagement and sales.

Q5. Dataset: Traffic Data

Question
Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

Answer
1. Problem Identification
Traffic congestion occurs when a large number of vehicles use a road while vehicle speed decreases and travel delays increase.
The dataset contains:

- Vehicle Count
- Average Speed
- Delay

The objective is to identify locations with similar traffic conditions and identify **congestion zones**.
2. Suitable Clustering Technique
K-Means clustering can be used for this problem.
The three numerical features are:

- Vehicle Count
- Average Speed
- Delay

Before clustering, the features should be **normalized**, because vehicle count and speed have different numerical scales.
3. Dataset

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

4. Analyze Traffic Patterns
Low Congestion Group
Locations:
L1, L3, L5
Characteristics:

- Vehicle count around 200–220.
- Speed around 28–32 km/h.
- Delay around 9–12 minutes.

These locations have relatively better traffic flow.

High Congestion Group
Locations:
L2, L4
Characteristics:

- Vehicle count = 500–600.
- Speed = 10–15 km/h.
- Delay = 25–30 minutes.

These locations show strong signs of congestion.
5. Possible Clusters

Cluster	Locations	Vehicle Count	Speed	Delay	Interpretation
Cluster 1	L1, L3, L5	Low	Higher	Low	Low congestion
Cluster 2	L2, L4	High	Low	High	Severe congestion

6. Clustering Process
Collect traffic data
↓
Select vehicle count, speed and delay
↓
Normalize features
↓
Choose K
↓

Apply K-Means
↓
Calculate cluster centroids
↓
Assign locations to clusters
↓
Interpret traffic conditions
↓
Identify congestion zones

7. Identifying Congestion

A cluster can be classified as a **congestion cluster** when it has:

- High vehicle count
- Low average speed
- High delay

For the given dataset:
L2 and L4 clearly satisfy these conditions.
Therefore:

8. Traffic Management Applications

1. Traffic signal optimization

Traffic authorities can adjust signal timings at highly congested locations.

2. Route diversion

Vehicles can be redirected through alternative roads.

3. Real-time traffic monitoring

Congestion clusters can be monitored continuously.

4. Infrastructure improvement

Locations that repeatedly appear in the high-congestion cluster may require:

- Road widening
- Additional lanes
- Improved intersections
- Better traffic signals

5. Emergency response

Traffic authorities can identify areas where emergency vehicles may face delays.

6. Intelligent transportation systems

Clustering results can be integrated with smart traffic systems to automatically detect changing congestion patterns.

9. Example Decision

Suppose L4 repeatedly appears in the severe-congestion cluster.
The traffic department could:

- Analyze peak-hour traffic.
- Adjust signal timing.
- Introduce alternative routes.
- Monitor vehicle flow.
- Consider infrastructure improvements.

10. Business and Public Benefits

Clustering-based traffic analysis can:

- Reduce travel time.
- Reduce fuel consumption.
- Reduce traffic delays.
- Improve road utilization.
- Improve emergency response.
- Reduce congestion-related pollution.
- Support smart-city planning.

Conclusion

K-Means clustering can group traffic locations according to vehicle count, average speed, and delay. In the given dataset, **L2 and L4 form a high-congestion cluster**, while **L1, L3, and L5 form a relatively low-congestion cluster**. This information can help traffic authorities optimize signals, plan alternative routes, improve infrastructure, and develop effective **intelligent traffic management systems**.

FINAL SUMMARY OF ALL 5 PROBLEMS

Q	Industry	Suitable Clustering	Main Clusters/Result	Main Application
Q1	Supermarket	K-Means	C1,C3,C5 and C2,C4	Targeted marketing

Q2	Banking/Fraud Detection	K-Means + anomaly detection	T1,T3,T5 normal; T2,T4 potential anomalies	Fraud monitoring
Q3	Telecom	K-Means	U1,U3,U5 high usage; U2,U4 low usage	Churn reduction
Q4	E-Commerce	K-Means	U1,U3,U5 high engagement; U2,U4 low engagement	Personalized recommendations
Q5	Transportation	K-Means	L1,L3,L5 low congestion; L2,L4 high congestion	Traffic management